Simulated Smart-Home Control with Small Language Models on SBCs: Task Success, Latency, and Replication

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***Abstract*—We evaluate a finite-rule baseline and quantized Qwen2.5 models of 0.5B and 1.5B parameters on Labrador single-board computers (SBCs). The home and every action are simulated; hardware and inference are real. Forty reserved scenarios cover clear instructions, paraphrases, explicit routines, ambiguity, and invalid requests, with two repetitions per condition. Scenario-averaged success is computed from the two repetitions, while latency statistics describe individual requests. Mean scenario success was 52.5% for rules, 37.5% for 0.5B, and 15.0% for 1.5B, with request medians 0.157, 36.848, and 94.663 s, respectively. Independent replicas on one, three, and six boards serve a fixed six-request batch. The released attempts distinguish model proposals from executor interventions and preserve unsuccessful executions. The scope supports conclusions about the tested configurations and constructed cases, without claiming general residential accuracy or maximum service capacity.**

***Index Terms*—simulated smart home, small language models, SBC, evaluation, latency**

I. Introduction and Research Question

Natural-language domestic control requires deciding which action to perform, when to request missing information, and when to reject an unsupported instruction. Structured output alone does not establish that the action is correct. On an SBC, an adequate decision must also be assessed against response time and memory cost. We therefore ask: how do a rule baseline and two small quantized models differ in task completion and latency, and how does the rate of correct responses in a fixed batch change with independent replicas?

HomeBench evaluates valid and invalid instructions for one or several devices in a virtual home [1]. Its coverage motivates separating incorrect proposals from syntactic validity. Lu et al. jointly examine small-model capabilities and deployment costs, including prefill, decoding, quantization, and hardware effects [2]. Our two-room Portuguese workload is a separate constructed dataset; it is neither a HomeBench replication nor evidence that findings from other edge hardware hold on Labrador. Qin et al. [3] study model selection and personalization under edge resource constraints, considering task difficulty, customization, compression, data, and learning time. Our focus is inference cost and simulated domestic decisions with fixed pretrained models. This motivates evaluating model, task, and resources jointly; our 0.5B/1.5B comparison neither replicates their personalization experiments nor causally confirms their guidelines. Our contributions are an auditable separation of proposed actions, executor intervention, and task outcomes; a joint comparison of a finite-rule baseline and two small-model configurations on ARM hardware; and a closed-batch replication study with its measurement limits made explicit. The study measures decision quality and deployment cost together, without proposing a new language model or a residential control system.

II. Architecture and Decision Boundaries

A Python coordinator constructs a fresh simulated state per request, calls a resident HTTP service, validates the returned JSON, applies permitted changes to Python dictionaries, and independently checks the outcome. No domestic device is connected to the executor. The domain has sala and quarto, lights with on/off values, single actions, an explicit two-step routine, clarification, and rejection. The schema admits `set_light`, `set_lights`, `ask_clarification`, and `no_action`. Plans are atomic: a permission failure in any step prevents the entire plan from being applied.

The executor receives only request, visible state, and permissions. It cannot consult expected states, prohibited proposals, scenario category, or other evaluator fields. A general conservative policy blocks lighting proposals when no known room name occurs as a whole word in the request and no valid location is supplied. Unknown or unauthorized rooms/values also block execution. Room evidence permits an incorrect-room proposal to reach the simulator; only the independent evaluator establishes correctness. Lexical room evidence does not validate intent in negative or unsupported requests. We record proposals, blocks, applied changes, and task outcomes separately.

Clarification requires requesting the missing room, preserving state, and producing no inappropriate action. Full-question conservative patterns recognize explicit requests such as “Qual cômodo?”. Merely mentioning a room word does not pass. Other questions enter a case-by-case review ledger; reviewer identity and rationale are retained, and unresolved questions

never automatically succeed. Assisted reviews are identified separately from human review.

III. Testbed and Frozen Protocol

Six Labrador boards report four ARM64 CPU cores, Debian 12, and approximately 1.89 GiB of OS-visible RAM. The same runtime binary and all shared libraries were verified by SHA-256. We use `llama-server` [4] build 9584/`e25a32e98`, CPU execution, four compute and four batch threads, context 512, and one inference slot per resident model. Both model servers remain loaded on each board, with at most one inference active across them. Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct [5], [6] are evaluated as local Q5_K_M GGUF files with preserved hashes. The hashes match the GGUF files published by Qwen byte for byte [7], [8]; model identity is thus established beyond filenames and metadata. The historical local download procedure is not preserved.

The rule baseline runs as a resident stateless Python HTTP service on the same boards. Its finite grammar recognizes acenda/ligue and apague/desligue, explicit room names, simple conjunctions, known-location commands, and edge-position politeness formulas. Unsupported forms return `no_action`. This grammar is a limited engineering reference, not an optimal natural-language rule system.

Eight development cases were used before freeze. A first compact prompt exposed unrequested lighting changes; a second development campaign tested general preservation instructions and the rule politeness correction. Both campaigns remain separate. Reserved outputs were unavailable during configuration selection. The freeze manifest versions scenarios, expected states and steps, prompt, schema, blocking policy, runtime/model settings, repetitions, and measurement scripts. Each inference uses a complete model on one board, with no model partitioning or agent collaboration.

The main dataset contains eight cases in each of five categories (40 distinct cases). Each condition has two repetitions per case ($n = 80$ requests; 240 total). All models run across all hosts, with repetitions of a case assigned to different boards. Each host processes a sequential queue; no board runs concurrent inferences in this comparison. Requests use compact general English instructions with Portuguese user text, temperature zero, seeds 801/802, 128 output tokens, and a 180 s timeout. Prompt cache is disabled and actual cache counts are retained. Temperature zero limits decision diversity; repetitions assess operational stability.

Success is conjunctive: normal operational completion (HTTP complete, generation stopped normally), contract-valid JSON, correct proposed action/steps, expected final state, preservation of unrelated state, and no prohibited or extra steps. Clear instructions and paraphrases require the requested light change with the other light preserved; one-step plans and single-light actions are equivalent. Routines require exactly both expected steps. Ambiguities require an informative question for the missing room and unchanged lights; invalid requests require `no_action` and unchanged lights.

TABLE I
Reserved attempts per condition ($n = 80$ requests, 40 scenarios). Latency in seconds; p95 by nearest rank. Undue and blocked columns count light commands in valid structured replies, not unique scenarios or complete plans.

Condition	Success	Median	p95	Undue	Blocked
Rules	42/80	0.157	0.643	0	0
Qwen 0.5B	30/80	36.848	45.292	36	16
Qwen 1.5B	12/80	94.663	159.117	76	22

An operational failure prevents normal completion; a format failure violates the output contract. Semantic/decision failures violate the action, clarification, rejection, or state criteria despite a valid reply. Executor intervention is a separate permission decision: blocking an incorrect proposal does not make the task successful. Conversely, `no_action` is correct for an invalid request but fails a valid command or a request needing clarification. Two repetitions are grouped by scenario; category averages give each scenario equal weight. No binomial precision interval or population-level significance claim treats repetitions as independent cases.

IV. Measurement

Coordinator monotonic time spans HTTP dispatch through response read and decode, including communication and inference and excluding model startup and SSH memory snapshots. Median and nearest-rank p95 are descriptive request statistics with sample sizes stated. Tokens and prompt/generation timings come from the runtime. Rules have no model tokens. RSS snapshots are before/after values; process VmHWM is a high-water mark since process start, not a per-request peak. RSS describes each process and may include shared mappings; it is not exclusive physical memory of the model. Other resident services are documented, so exclusive board memory is not asserted.

Scale tests use independent replicas with one request per board per wave, one, three, or six active boards, and two six-request batches per size/condition (108 total requests). The same six cases include one clear command, one paraphrase, two routines, one ambiguity, and one invalid request. The batch window runs from first dispatch to last response, including instrumented gaps between waves. We separately report individual latency, responses/min, and correctly completed tasks/min. Correct-task rate uses the number judged successful over this HTTP response window; final simulation application and grading occur after receipt. It is a rate of correct task responses, not measured time until a physical or simulated state change. This finite closed workload does not establish saturation throughput, queued-arrival latency, or maximum service capacity. Each request starts from a fresh state with no conversation history.

TABLE II

SUCCESS BY CATEGORY: EIGHT SCENARIOS WITH TWO REPETITIONS (16 ATTEMPTS) PER CELL. PER-SCENARIO FRACTIONS ARE IN THE REPRODUCIBLE CSV. *FOUR 0.5B AMBIGUITY JUDGMENTS AWAIT HUMAN VALIDATION; NO ASSISTED SUCCESS IS AWARDED.

Category	Rules	0.5B	1.5B
Clear instructions	16/16	8/16	0/16
Paraphrases	0/16	14/16	4/16
Routines	8/16	8/16	6/16
Ambiguity	2/16	0/16*	0/16
Invalid requests	16/16	0/16	2/16

TABLE III

FAILED REQUIREMENTS BY ATTEMPT (80 PER CONDITION); OVERLAPPING ROWS MUST NOT BE SUMMED. PARENTHESES GIVE APPLICABLE/OBSERVED ATTEMPTS: STEPS AND TARGET STATE CONCERN CLEAR/PARAPHRASE/ROUTINE, PRESERVATION EXCLUDES ROUTINES; p DENOTES PROPOSED REPLIES OF THAT TYPE. PROHIBITION CHECKS ARE RECORDED IN ALL 80, WITH NONEMPTY PROHIBITIONS ONLY IN 32 AMBIGUITY/INVALID ATTEMPTS. *FOUR FAILURES AWAIT HUMAN VALIDATION; –/0 MEANS NO PROPOSAL, NOT DEMONSTRATED ABILITY.

Requirement (denominator)	Rules	0.5B	1.5B
Wrong step set (48)	24	18	38
Wrong target state (48)	24	18	22
Preservation failed (64)	0	8	14
Undue `no_action` (p)	38	4	20
Proposal blocked (80)	0	14	12
Prohibited step (80)	0	26	24
Inadequate clarification (p)	0/2	12/12*	–/0

V. RESULTS

A. Reserved quality and latency

All 240 replies were structurally valid and completed normally; schema validity therefore did not establish correct control. Rules completed 42/80 attempts, Qwen 0.5B 30/80, and Qwen 1.5B 12/80, all credited automatically. Rules completed all 16 clear and 16 invalid attempts but none of the 16 paraphrases. The 0.5B configuration completed 14/16 paraphrases but rejected none of the invalid requests correctly. Neither model completed an ambiguity case under the recorded criteria; four 0.5B judgments remain provisional. Category fractions group two repetitions of each of eight scenarios, not 16 independent scenarios.

Four 0.5B clarification responses have assisted failure judgments awaiting human validation in the preserved ledger. They are provisionally unsuccessful; no reported success is awarded by assisted review. Even crediting all four would raise 0.5B to 34/80 (42.5%), below rules at 42/80 (52.5%). This sensitivity bound does not replace semantic review.

At command level, 0.5B proposed 36 undue light changes, 16 blocked and 20 applied only in simulation; 1.5B proposed 76, 22 blocked and 54 simulated. These counts differ from reply-level diagnostics, which can also include omitted steps. Median post-request RSS was 571.6 MiB for 0.5B and 1183.7 MiB for 1.5B. Both used 124–138 prompt tokens; output ranges were 16–40 and 12–75 tokens. Recorded prompt-cache counts were zero throughout.

B. Failure criteria and illustrative cases

A complementary post-hoc analysis groups existing evaluator checks, counting each group at most once per attempt. It does not change the frozen evaluator or outcomes. Table III reports failed requirements rather than exclusive causes: state, step, and permission checks can fail together. The artifact records exact check names, prefix mappings, observed/pass/fail/absent counts, and attempt paths. An absent check is not a pass; for example, clarification is checked only when proposed. The complete decomposition includes expected-action errors in ambiguity/invalid cases, while step equality handles lighting wrappers. Rules' 38 unsuccessful attempts all returned `no_action` outside the accepted rejection policy: 16 paraphrases, 8 routines, and 14 ambiguities. They preserved unrelated state but omitted required actions or clarification. The 0.5B configuration instead proposed prohibited light steps in 14/16 invalid attempts and inappropriate clarification in the other 2; 8 invalid attempts changed protected state. For 1.5B, all 16/16 clear attempts had incorrect step sets, but only 8 failed the target-state check: a correct target value alone can conceal an extra change. Its 38 wrong-step attempts and 14 preservation failures characterize this configuration, without establishing why it behaved this way.

Clarifications proposed/correct were 2/2, 12/0, and 0/0 for rules, 0.5B, and 1.5B. All 12 proposed 0.5B clarifications failed the question criterion; 4 are provisional assisted judgments. The 1.5B model proposed none; zero clarification-check failures would not demonstrate clarification ability. The p denominators for proposed `no_action` are 54, 4, and 22.

The following examples illustrate recorded behavior, not a causal model or a substitute for the counts. IDs are attempt-directory names under the quality run; full JSON, initial/expected/final states, failed checks, and source paths are in the complementary artifact. Portuguese request and output excerpts are retained verbatim; action notation abbreviates recorded JSON fields.

E1 (`rules-held-paraphrase-01-rep1-169`).

"Por favor, deixe a luz da sala ligada." expects sala=on and quarto=on unchanged. Rules returned `no_action` with reason "Pedido fora das regras explícitas.". No block occurred; sala stayed off. Final-state, exact-step, and no-action checks failed.

E2 (`05-held-ambiguity-01-rep1-169`).

"Ligue a luz, por favor.", with no known location, requires asking the room. The 0.5B reply was `set_light`, `room=sala`, `value=on`. The executor blocked it for insufficient room information; both lights remained unchanged (sala=off, quarto=on). Permission, expected-action, and prohibited-step checks failed despite preservation.

TABLE IV
TWO SIX-REQUEST BATCHES PER CONDITION/SIZE. RATE IS MEAN CORRECT RESPONSES/MIN IN THE INSTRUMENTED BATCH WINDOW, NOT SUSTAINED CAPACITY; INDIVIDUAL MEDIAN AND P95 USE $n = 12$ REQUESTS. BATCH RANGES AND RESPONSES/MIN ARE IN THE ARTIFACT.

Condition	Boards	Success	Rate	Median	p95
Rules	1	10/12	42.753	0.006	0.211
Rules	3	10/12	235.196	0.006	0.008
Rules	6	10/12	30555.536	0.006	0.011
Qwen 0.5B	1	6/12	0.871	24.586	37.903
Qwen 0.5B	3	6/12	2.883	23.993	40.917
Qwen 0.5B	6	6/12	6.393	23.990	28.772
Qwen 1.5B	1	2/12	0.085	84.187	131.203
Qwen 1.5B	3	2/12	0.289	78.606	139.278
Qwen 1.5B	6	2/12	0.661	80.019	93.518

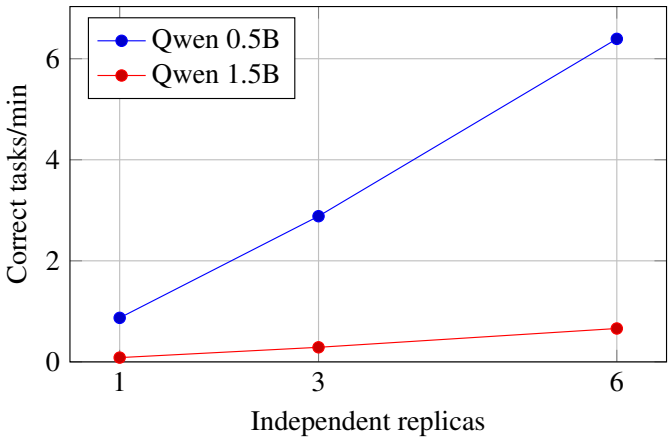


Fig. 1. Correct tasks/min averaged over two instrumented closed batches. Individual latency and unsuccessful decisions are tabulated; curves do not demonstrate maximum capacity.

E3 (`05-held-ambiguity-03-rep1-240`).
"Quero a luz acesa.", with no known location, also requires asking the room. The 0.5B returned `ask_clarification`, `missing_field=room`, with question "Quero a luz acender.". No block or state change occurred (sala=off, quarto=on). Only the informative-question check failed provisionally: the recorded Codex judgment awaits human validation.
E4 (`15-held-clear-01-rep1-240`).
"Ligue a luz na sala." expects only sala=on, preserving quarto=on. The 1.5B returned a `set_lights` plan with `(sala,on)` and `(quarto,off)`. Both commands were permitted and simulated; final sala=on, quarto=off. Exact-step and preservation checks failed, although the target-light check passed. One reply contains two commands, including one extra change.

C. Replicated service

The batch CSV preserves each window, successful-task rate, and response rate separately. Table IV pools twelve individual requests per condition/size for latency and averages two batch rates; ranges remain in the machine-readable summary.

For 0.5B, mean correct-task response rates were 0.871, 2.883, and 6.393/min with one, three, and six replicas; 1.5B rates were 0.085, 0.289, and 0.661/min. Each six-request batch retained three correct tasks for 0.5B, one for 1.5B, and five for rules. The rule rate at six replicas reflects a response window of about 9–11 ms and removal of between-wave instrumentation gaps, so it must not be interpreted as measured sustained service capacity.

VI. DISCUSSION AND LIMITATIONS

Task completion and schema validity are separate outcomes. A blocked incorrect proposal remains an incorrect model decision, and higher response volume cannot repair that decision. Replication changes service concurrency, while the model and simulator contract remain fixed. The failure profile explains what aggregate success hides: the 0.5B configuration handles more of these paraphrases than rules, yet loses on invalid-request rejection and state preservation. The 1.5B configuration has higher latency and more wrong-step proposals in this setup; neither model size nor an internal reasoning mechanism was isolated experimentally. LoRA, RAG, or task specialization would require new development data and a separate protocol; none was implemented here. Failed simulated decisions establish neither safety nor unsafety in a real home. The 40 cases are authored utterance/state combinations rather than 40 different homes; several express the same underlying action. Requested lights initially have the opposite value, so idempotent requests and alternative state distributions are untested. Routine success alone cannot establish language understanding independent of a state-toggling strategy. The workload is small, Portuguese, and restricted to two light controls. The rule grammar is deliberately finite, the JSON contract constrains generation, and one quantization is used per size. This is a configuration comparison, not an isolated parameter-count or architecture effect. Interleaving reduces model/host confounding without eliminating thermal, memory, and network variation. Services remain resident, but disk/page caches are not cleared. Memory snapshots do not measure per-request peaks, and no energy measurement is made. Scale batches are few and include coordinator instrumentation costs, which can dominate the rule baseline. Four assisted clarification judgments remain provisional; their possible effect is bounded above. The simulator cannot establish real-home safety or usability.

VII. CONCLUSION AND NEXT STEPS

In the reserved workload, rules completed 42/80 attempts, the 0.5B configuration 30/80, and the 1.5B configuration 12/80; all replies were structurally valid. The 0.5B configuration covered 14/16 paraphrases but rejected none of the invalid requests correctly. These outcomes separate decision quality from structural validity and operational intervention. The measured counts and latency characterize the frozen configurations on these constructed cases. Further work should expand independently reviewed cases and real-user phrasing before any residential deployment. Collaboration among agents and cache comparisons require separate future protocols.

Evidence Availability and AI-use Disclosure

The complementary failure analysis, reproduction command, and attempt-linked examples are available on the revision branch; its input manifest links every source to the original scientific evidence. The frozen protocol, all 240 quality and 108 replication attempts, raw responses, unsuccessful outcomes, review ledger, and offline reconstruction scripts are public at an immutable repository snapshot [9]. Reproduction checks recorded state, permissions, and evaluator outputs against every response; the supplied scripts regenerate the tables and plot sources. Re-running hardware additionally requires the recorded runtime and model files; offline reconstruction does not constitute an independent hardware replication.

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